MANUAL MAESTRO DE CONFIGURACIÓN PASO A PASO: pfSense CE 2.9.0

BLUEPRINT VISUAL, SIMULACIÓN WebGUI Y PARÁMETROS OFICIALES DE LABORATORIO

Autor: Bruno Urrea Ortiz | Duoc UC Sede San Joaquín • **Portafolio de Título APT122** ($0 CLP)

FASE 1: SETUP INICIAL, HARDWARE TUNING & NETWORKING BASE

PASO 1.1: CONFIGURACIÓN GENERAL DEL SISTEMA Y SERVIDORES DNS

■ **pfSense WebGUI** // **System / General Setup** v2.9.0-RELEASE (amd64)	
Hostname & Domain	kronos-fw . kronos.local
DNS Server Settings	DNS 1: 1.1.1.1 (Gateway: none) | DNS 2: 8.8.8.8
DNS Server Override	[ ] Allow DNS server list to be overridden by DHCP on WAN
DNS Resolution Behavior	Use local DNS (127.0.0.1), fall back to remote DNS Servers [▼]
Timezone & NTP	America/Santiago (Chile Standard Time) | 0.south-america.pool.ntp.org
Acciones WebGUI	**[Save] [Apply Changes]**

PASO 1.2: HARDWARE OFFLOADING TUNING (MANDATORIO PARA NETMAP INLINE IPS)

■ **pfSense WebGUI** // **System / Advanced / Networking ➔ Network Interfaces** v2.9.0-RELEASE (amd64)	
Hardware Checksum Offload	[X] Disable hardware checksum offload (Desactiva cálculo por NIC)
Hardware TCP Segmentation	[X] Disable hardware TCP segmentation offload (TSO) (Mandatorio para Netmap)
Hardware Large Receive	[X] Disable hardware large receive offload (LRO) (Evita agregación previa a IPS)
Netgate Doc Note	*CRÍTICO: El framework netmap(4) en modo Inline IPS colisiona si el hardware offloading está activo.*
Acciones WebGUI	**[Save] [Apply Changes] (Requiere reiniciar el sistema)**

PASO 1.3: CONFIGURACIÓN DE TRONCAL 802.1Q Y CREACIÓN DE VLANs

Navegar a Interfaces > Assignments > VLANs y registrar los 4 tags de subred sobre el adaptador físico:

■ pfSense WebGUI // Interfaces / VLANs / Edit	v2.9.0-RELEASE (amd64)
Parent Interface	vtnet1 (LAN Trunk Physical Adapter) [▼]
VLAN Tag 10 (Corporativa)	Tag: 10 | Priority: 0 | Description: VLAN_10_CORP_INTERNAL
VLAN Tag 20 (DMZ Web)	Tag: 20 | Priority: 0 | Description: VLAN_20_DMZ_SERVERS
VLAN Tag 30 (VoIP PBX)	Tag: 30 | Priority: 0 | Description: VLAN_30_VOIP_PBX
VLAN Tag 99 (Gestión SecOps)	Tag: 99 | Priority: 0 | Description: VLAN_99_MGMT_SEC
Acciones WebGUI	**[Save] [Apply Changes]**

PASO 1.4: ASIGNACIÓN DE INTERFACES Y DIRECCIONAMIENTO IP ESTÁTICO

Navegar a Interfaces > Assignments para enlazar cada VLAN y configurar su puerta de enlace:

■ pfSense WebGUI // Interfaces / DMZ_SERVERS (vtnet1.20) & WAN (vtnet0)	v2.9.0-RELEASE (amd64)
Enable Interface	[X] Enable interface
IPv4 Configuration Type	Static IPv4 [▼]
IPv4 Address (DMZ_SERVERS)	192.168.20.1 / 24 | IPv4 Upstream Gateway: None [▼]
IPv4 Address (LAN_CORP)	192.168.10.1 / 24 | IPv4 Address (VOIP_PBX): 192.168.30.1 / 24
IPv4 Address (MGMT_SEC)	192.168.99.1 / 24 | IPv4 Address (WAN): 198.51.100.1 / 24 (Lab Dual-Host)
WAN RFC 1918 / Bogon Filter	[ ] Block private networks [ ] Block bogon networks (Desmarcados en Lab)
Acciones WebGUI	**[Save] [Apply Changes]**

FASE 2: SERVICIOS DHCP Y RESERVAS ESTÁTICAS DE INFRAESTRUCTURA

PASO 2.1: SERVIDOR DHCP Y RESERVA FIJA PARA EL SERVIDOR WEB DMZ (DVWA)

■ **pfSense WebGUI** // **Services / DHCP Server / DMZ_SERVERS (VLAN 20)**v2.9.0-RELEASE (amd64)	
Enable DHCP Server	[X] Enable DHCP server on DMZ_SERVERS interface
Subnet & Subnet Mask	192.168.20.0 / 255.255.255.0 (Available Range: .1 to .254)
Dynamic Pool Range	From: 192.168.20.100 To: 192.168.20.199
DNS & Default Gateway	DNS: 192.168.20.1, 1.1.1.1 | Gateway: 192.168.20.1
Static Mapping DVWA Target	MAC: 02:42:c0:a8:14:32 ➔ IP Fija: 192.168.20.50 (dvwa-dmz-target)
Static Lease Description	Laboratorio Web Vulnerable Controlado - DMZ
Acciones WebGUI	**[Save] [Apply Changes]**

PASO 2.2: SERVIDOR DHCP Y RESERVA FIJA PARA CENTRALITA ASTERISK PBX

■ **pfSense WebGUI** // **Services / DHCP Server / VOIP_PBX (VLAN 30)**v2.9.0-RELEASE (amd64)	
Enable DHCP Server	[X] Enable DHCP server on VOIP_PBX interface
Dynamic Pool Range	From: 192.168.30.100 To: 192.168.30.199
DNS & Default Gateway	DNS: 192.168.30.1 | Gateway: 192.168.30.1
Static Mapping Asterisk PBX	MAC: 02:42:c0:a8:1e:32 ➔ IP Fija: 192.168.30.50 (asterisk-pbx-core)
Static Lease Description	Centralita Telefonica SIP/PJSIP y Auto-Dialer SOAR
Acciones WebGUI	**[Save] [Apply Changes]**

FASE 3: SURICATA 7.X — PREVENCIÓN EN KERNEL (NETMAP INLINE IPS)

PASO 3.1: CONFIGURACIÓN GLOBAL DE REGLAS (ET OPEN & SNORT COMMUNITY)

■ pfSense WebGUI // **Services / Suricata / Global Settings** v2.9.0-RELEASE (amd64)	
Install ETOpen Rules	[X] Install Emerging Threats Open rules (Free Community Edition $0)
Install Snort Rules	[X] Install Snort Community rules (Cisco Talos Open Ruleset $0)
Snort Oinkmaster Code	none (No requerido para Snort Community Free Tier)
Update Interval & Live Swap	Update Interval: 12 Hours [▼] | [X] Live rule swap on update
Acciones WebGUI	**[Save]**

PASO 3.2: CONFIGURACIÓN DE LA INTERFAZ WAN EN MODO INLINE IPS (NETMAP)

■ pfSense WebGUI // **Services / Suricata / Interfaces / Edit [WAN Settings]** v2.9.0-RELEASE (amd64)	
Enable & Interface	[X] Enable inspection | Interface: WAN (vtnet0) [▼]
IPS Mode Selection	( ) Legacy Mode (X) Inline Mode (netmap framework)
Block on Alerts & Kill States	[X] Block offenders (Hardware ring-buffer drop) | [X] Kill active states
EVE JSON Telemetry	[X] Enable EVE JSON Log | Type: (X) FILE (/var/log/suricata/eve.json)
EVE Event Types	[X] Alerts (SQLi/RCE) [X] HTTP (URI, Host, Headers) [X] TLS
Acciones WebGUI	**[Save] [Apply Changes]**

PASO 3.3: GESTIÓN DE FIRMAS ET OPEN Y POLÍTICA dropsid.conf

■ pfSense WebGUI // **Services / Suricata / SID Mgmt / dropsid.conf** v2.9.0-RELEASE (amd64)	
Automatic SID Mgmt	[X] Enable Automatic SID State Management
Drop SID List Selection	dropsid.conf [▼] (Transforma alertas en Drops atómicos)
Categorias PCRE Monitoreadas	pcre:emerging-sql, pcre:emerging-exploit, pcre:community-web-attacks
State Order Evaluation	Drop > Enable > Disable
Acciones WebGUI	**[Save] [Apply Changes]**

FASE 4: INTELIGENCIA DE AMENAZAS CON pfBlockerNG-devel Y MAXMIND GEOIP

PASO 4.1: CONFIGURACIÓN DE MAXMIND GEOIP FREE TIER

■ pfSense WebGUI // Firewall / pfBlockerNG / IP / GeoIPv2.9.0-RELEASE (amd64)	
MaxMind Account ID & Key	Account ID: 1024982 | License Key: ******************** ($0 Free Tier)
Top Spammers GeoIP	CN, RU, IR, KP, VN, NG [▼]
List Action	( ) Disabled ( ) Permit (X) Deny Inbound ( ) Deny Both
Logging Configuration	[X] Enable pfBlockerNG log (/var/log/pfblockerng/ip_block.log)
Acciones WebGUI	**[Save] [Apply Changes]**

PASO 4.2: FEEDS DE REPUTACIÓN IP GLOBALES (FireHOL & Spamhaus DROP)

■ pfSense WebGUI // Firewall / pfBlockerNG / IP / IP Feeds / Editv2.9.0-RELEASE (amd64)	
Feed 1 (FireHOL L1)	URL: https://iplists.firehol.org/files/firehol_level1.netset | Action: Deny Inbound
Feed 2 (Spamhaus DROP)	URL: https://www.spamhaus.org/drop/drop.txt | Action: Deny Inbound
Update Frequency	Every 4 Hours [▼] | State: ON [▼]
Automatic Rule Order	| pfB_Block (Floating Rules Top Priority) [▼]
Acciones WebGUI	**[Save] [Apply Changes]**

FASE 5: PROXY INVERSO HAPROXY 2.8+ Y PROTECCIÓN ANTI-DOS / FUZZING

PASO 5.1: CONFIGURACIÓN DEL BACKEND (DVWA EN DMZ)

■ pfSense WebGUI // Services / HAProxy / Backend / Edit: DVWA_DMZ_POOL	v2.9.0-RELEASE (amd64)
Backend Name & Mode	Name: DVWA_DMZ_POOL | Mode: active [▼] | Balance: roundrobin [▼]
Server List Target	dvwa-node → Forward to: Address+Port → IP: 192.168.20.50 | Port: 80
Backend Encryption	[ ] Encrypt connection to backend (HTTP Plano en DMZ interna)
Health Check Method	HTTP [▼] | HTTP check URI: /index.php | Check Frequency: 2000 ms
Acciones WebGUI	**[Save] [Apply Changes]**

PASO 5.2: CONFIGURACIÓN DEL FRONTEND SSL Y STICK-TABLES DE RATE LIMITING

■ pfSense WebGUI // Services / HAProxy / Frontend / Edit: KRONOS_HTTPS_VIP	v2.9.0-RELEASE (amd64)
Frontend Name & Address	Name: KRONOS_HTTPS_FRONTEND | Listen: WAN address (198.51.100.1):443
SSL Offloading Cert	Type: (X) SSL / HTTPS offloading | Cert: KRONOS_LAB_CERT (RSA 2048)
Advanced Pass Thru (Stick Tables)	stick-table type ip size 100k expire 30s store http_req_rate(10s),http_err_rate(10s)
Anti-DoS ACL Rule 1	http-request track-sc0 src | http-request deny deny_status 429 if { sc_http_req_rate(0) gt 100 }
Anti-Fuzzing ACL Rule 2	http-request deny deny_status 403 if { sc_http_err_rate(0) gt 25 }
Default Backend Pool	DVWA_DMZ_POOL [▼]
Acciones WebGUI	**[Save] [Apply Changes]**

FASE 6: MATRIZ DE REGLAS DE FIREWALL ZERO TRUST

PASO 6.1: REGLAS DE AISLAMIENTO DMZ Y PERÍMETRO WAN

■ **pfSense WebGUI** // **Firewall / Rules / DMZ_SERVERS (VLAN 20)**v2.9.0-RELEASE (amd64)	
Regla 1 (DNS/NTP Base)	PASS IPv4 UDP ➔ Source: DMZ_SERVERS ➔ Dest: * : Ports 53, 123
Regla 2 (Aislamiento Corp)	DROP IPv4 * ➔ Source: DMZ_SERVERS ➔ Dest: LAN_CORP (192.168.10.0/24)
Regla 3 (Aislamiento Mgmt)	DROP IPv4 * ➔ Source: DMZ_SERVERS ➔ Dest: MGMT_SEC (192.168.99.0/24)
Regla 4 (Salida Web Updates)	PASS IPv4 TCP ➔ Source: DMZ_SERVERS ➔ Dest: * : Ports 80, 443
Regla WAN (Acceso VIP)	PASS IPv4 TCP ➔ Source: * ➔ Dest: WAN address : Port 443 (HAProxy VIP)
Acciones WebGUI	**[Save] [Apply Changes]**

FASE 7: ENLACE ZERO TRUST CON TAILSCALE SUBNET ROUTER (VOIP PBX)

■ **pfSense WebGUI** // **VPN / Tailscale / Settings**v2.9.0-RELEASE (amd64)	
Enable Tailscale Daemon	[X] Enable Tailscale service daemon
Authentication Key	Auth Key: tskey-auth-k98a-********************************
Advertised Subnet Routes	192.168.30.0/24 (Publica subred VoIP hacia la Tailnet privada)
Accept Subnet Routes	[X] Enable route passing for connected Softphones (Zoiper/Linphone)
Tailscale Admin Action	*En login.tailscale.com > Machines > pfSense > Edit route settings > Aprobar 192.168.30.0/24*
Acciones WebGUI	**[Save] [Apply Changes]**

FASE 8: VERIFICACIÓN Y DIAGNÓSTICO EN CONSOLA FREEBSD

Para validar la operatividad total del firewall, acceder vía SSH (ssh admin@192.168.99.1):

```
# 1. Verificar Suricata activo con hilos Netmap en tiempo real:
ps aux | grep suricata
netstat -i

# 2. Tail en vivo del log de eventos EVE JSON estructurado con jq:
tail -f /var/log/suricata/eve.json | jq '{time: .timestamp, src: .src_ip, dst: .dest_ip, alert: .alert.signature}'

# 3. Comprobar la tabla de persistencia en kernel de FreeBSD (snort2c):
pfctl -t snort2c -T show

# 4. Probar atómicamente si una IP atacante está bloqueada en memoria RAM:
pfctl -t snort2c -T test 198.51.100.100

# 5. Inserción manual de prueba en la tabla de kernel:
pfctl -t snort2c -T add 198.51.100.100

# 6. Eliminar estados de conexión residuales de la IP atacante:
pfctl -k 198.51.100.100

# 7. Comprobar estado de sockets y puertos en escucha (HAProxy y Syslog):
sockstat -4 -l
```

CONCLUSIÓN DEL DESPLIEGUE: Siguiendo este blueprint paso a paso, el firewall pfSense CE 2.9.0 queda configurado con hardening perimetral de nivel profesional, inspección de paquetes en microsegundos sin falsos positivos y orquestación de respuesta autónoma SOAR.